

MATH-361 Probability & Statistics

DE-45-CE-A

Project Total Marks = 50

Group-1 and Group-2

Title

Sampling-Based Detection of Network Anomalies Using Statistical Methods

Problem Statement

Modern computer networks generate massive amounts of traffic data, making it inefficient and computationally expensive to analyze every packet. Security systems often rely on analyzing partial data, but poor sampling can miss important anomalies (e.g., cyber attacks).

The problem is:

How can we design an efficient sampling method that reduces data size while still accurately detecting anomalous or malicious network behavior using statistical techniques?

What You Are Required To Do

In this project, you will design and evaluate a sampling-based approach for anomaly detection.

You should begin by clearly defining what you consider an anomaly in your dataset. This definition must be supported using statistical reasoning for example, identifying outliers based on distributions, thresholds, or deviations from normal behavior.

Next, you will select an appropriate dataset. You may use a standard network intrusion dataset, or you may generate your own synthetic dataset containing both normal and anomalous behavior. In either case, you must clearly explain the nature of your data.

The central part of the project is the design of your sampling method. You may use a standard technique such as random, stratified, or systematic sampling, or you may propose an improved or adaptive method. Your method must be explained clearly, including its working steps and the probability-based reasoning behind it. You should also explain why your approach is expected to preserve anomalies.

You are required to apply statistical techniques to analyze your data. This includes using concepts such as probability distributions, mean and variance, or hypothesis testing.

These should not only be implemented in code but also explained mathematically and interpreted properly.

After this, you will perform anomaly detection on both the full dataset and the sampled dataset. A key part of your work is comparing these two cases to evaluate how much information is lost due to sampling and how it affects detection performance.

Your evaluation should include measures such as detection accuracy, false positives, and the percentage of data reduction. Graphical representation of results is essential for clear comparison.

Finally, you must include a flowchart that shows the complete process—from data input to sampling, analysis, anomaly detection, and output.

What You Must Submit (in soft form)

You are required to submit a report in word format. The report should include an introduction, a clear definition of the problem, a detailed explanation of your methodology, results with proper discussion, and a conclusion.

In addition, you must submit your Python/MATLAB code. The code should be clean, properly structured, and well-commented so that your results can be reproduced.

Your results must include multiple graphs and at least one comparison table that clearly shows the difference between full-data and sampled-data performance.